Nondimensionalization and similarity

Fluid, transport, and immersed-structure parameters

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IBFE Tutorials

$$\mathrm{Re} = \frac{\rho U L}{\mu}, \qquad \mathrm{St}_L = \frac{fL}{U}$$

$$\alpha = L\sqrt{\frac{\omega}{\nu}}, \qquad \mathrm{Pe} = \frac{UL}{D}$$

AI tools were used to assist with formatting, organization, and clarification of the instructor's original notes.

Why nondimensionalize?

Nondimensional variables let us:

- compare systems with different sizes, speeds, and materials;
- identify which physical effects dominate;
- reduce the number of independent parameters;
- design dynamically similar experiments and simulations.

A nondimensional solution can always be converted back to physical units once the reference scales are known.

Choose and report the scales. A value such as $\mathrm{Re} = 100$ is only reproducible when L and U are defined.

Characteristic scales

Choose scales from the physics of the problem:

Quantity	Scale	Possible choice
Length	L	chord, diameter, body length
Velocity	U	mean inflow, swim speed, tip speed
Time	$T = L/U$	advective time
Pressure	$P_I = \rho U^2$	inertial pressure

Use the same L and U in every dimensionless group and coefficient unless a different scale is explicitly introduced.

The computational domain must be large enough that moving the outer boundary farther away does not change the quantity of interest. There is no universal "$10L$" rule.

Dimensionless variables and derivatives

Define starred variables by

$$\boldsymbol{x} = L\boldsymbol{x}^*, \qquad \boldsymbol{u} = U\boldsymbol{u}^*, \qquad t = \frac{L}{U}t^*, \qquad p - p_0 = \rho U^2 p_l^*.$$

The differential operators transform as

$$\nabla = \frac{1}{L}\nabla^*, \qquad \frac{\partial}{\partial t} = \frac{U}{L}\frac{\partial}{\partial t^*}, \qquad \frac{D\boldsymbol{u}}{Dt} = \frac{U^2}{L}\left[\frac{\partial \boldsymbol{u}^*}{\partial t^*} + (\boldsymbol{u}^*\cdot\nabla^*)\boldsymbol{u}^*\right].$$

A prime can be mistaken for a derivative. Stars make the dimensionless variables visually distinct throughout the derivation.

Inertial form of incompressible Navier–Stokes

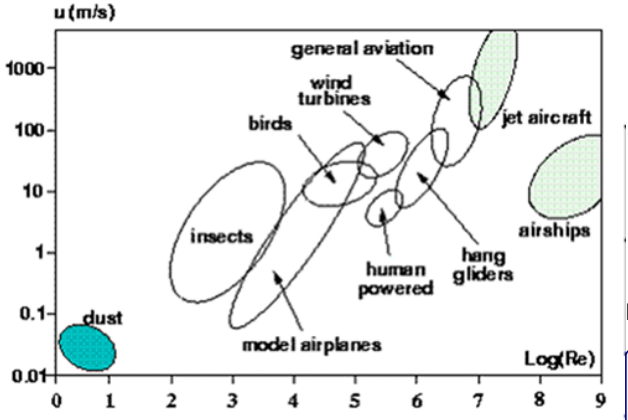
For a Newtonian fluid with constant density ρ and viscosity μ,

$$\boxed{\frac{\partial \boldsymbol{u}^*}{\partial t^*} + (\boldsymbol{u}^* \cdot \nabla^*)\boldsymbol{u}^* = -\nabla^* p_I^* + \frac{1}{\mathrm{Re}} \nabla^{*2} \boldsymbol{u}^* + \boldsymbol{b}^*}$$

$$\boxed{\nabla^* \cdot \boldsymbol{u}^* = 0}, \qquad \mathrm{Re} = \frac{\rho U L}{\mu} = \frac{U L}{\nu}, \qquad \nu = \frac{\mu}{\rho}.$$

Here $\boldsymbol{b}^* = (L/U^2)\boldsymbol{b}$ is a dimensionless body-force acceleration. If $\boldsymbol{b} = -g\hat{\boldsymbol{\jmath}}$, then $\boldsymbol{b}^* = -(1/\mathrm{Fr}^2)\hat{\boldsymbol{\jmath}}$, where $\mathrm{Fr} = U/\sqrt{gL}$ is the Froude number.

Reynolds number



Flight speed versus $\log_{10} \mathrm{Re}$, from dust to airships. Redrawn after P. B. S. Lissaman, *Annu. Rev. Fluid Mech.* 15 (1983), Fig. 1.

$$\mathrm{Re} = \frac{\text{inertial scale}}{\text{viscous scale}} = \frac{\rho U^2/L}{\mu U/L^2}.$$

- $\mathrm{Re} \ll 1$: viscous stresses dominate; inertia is a perturbation.
- $\mathrm{Re} \gg 1$: inertia is important; thin shear layers and separation may occur.

Each unit of the abscissa spans a decade in Re.

Reynolds number identifies a balance. It does not, by itself, determine whether a flow is turbulent.

Viscous normalization at low Reynolds number

The inertial pressure scale is convenient when inertia is important:

$$\frac{D\boldsymbol{u}^*}{Dt^*} = -\nabla^* p_I^* + \frac{1}{\mathrm{Re}}\nabla^{*2}\boldsymbol{u}^* + \boldsymbol{b}^*.$$

For creeping-flow problems, choose the viscous pressure scale

$$p - p_0 = \frac{\mu U}{L} p_V^*.$$

The same dimensional equation then becomes

$$\boxed{\mathrm{Re}\frac{D\boldsymbol{u}^*}{Dt^*} = -\nabla^* p_V^* + \nabla^{*2}\boldsymbol{u}^* + \mathrm{Re}\,\boldsymbol{b}^*}.$$

Both forms are valid at every Reynolds number. They satisfy $p_V^* = \mathrm{Re}\, p_I^*$, and the inertially scaled body acceleration acquires the same factor; there is no mathematical switch at $\mathrm{Re} = 1$.

Dimensionless time and frequency

The advective time scale L/U gives

$$t^* = \frac{tU}{L}, \qquad f^* = \frac{fL}{U}, \qquad \omega^* = \frac{\omega L}{U} = 2\pi f^*.$$

- One dimensionless cycle has period $1/f^*$.
- Matching f^* preserves the ratio of oscillation time to advection time.
- Pumping, flapping, and pulsatile-flow comparisons must use the same definitions of L and U.

A dimensional frequency cannot be compared across geometrically or dynamically different systems until it is combined with the relevant time scale.

Frequency scaling and the Strouhal number

For vortex shedding or another oscillation with frequency f,

$$\boxed{\mathrm{St}_L = \frac{fL}{U}}.$$

This compares the oscillation time scale $1/f$ with the advective time scale L/U.
For an oscillating wake, wing, or propulsor,

$$\boxed{\mathrm{St}_A = \frac{fA}{U}},$$

where A is the stated excursion scale. If $A = L$, then $\mathrm{St}_A = \mathrm{St}_L$.

State whether the characteristic length is a chord, diameter, wake width, or motion amplitude. Different choices define different Strouhal numbers.

Strouhal number in swimming and flying

For a swimming or flying animal, a common kinematic definition is

$$\boxed{\mathrm{St}_A = \frac{fA}{U}},$$

where f is stroke or tail-beat frequency, A is a stated excursion amplitude, and U is mean forward speed.

- St_A organizes vortex-growth and shedding regimes for pitching and heaving propulsors.
- Many cruising swimmers and flyers in the Taylor et al. data lie in $0.2 < \mathrm{St}_A < 0.4$.
- The interval is empirical, not a universal efficiency optimum for every body, gait, or amplitude convention.

State whether A is peak or peak-to-peak excursion. Taylor, Nudds, and Thomas, *Nature* 425 (2003).

Womersley number

For oscillatory viscous flow with angular frequency ω,

$$\boxed{\alpha = L\sqrt{\frac{\omega}{\nu}}}.$$

- $\alpha \ll 1$: viscosity communicates the oscillation across the length scale during each cycle.
- $\alpha \gg 1$: unsteady inertia dominates away from a thin oscillatory boundary layer.

$$\delta_\omega \sim \sqrt{\frac{2\nu}{\omega}}, \qquad \alpha = \sqrt{2}\,\frac{L}{\delta_\omega}.$$

If Re and $\mathrm{St}_L = fL/U$ use the same L and U,

$$\boxed{\alpha^2 = 2\pi\,\mathrm{Re}\,\mathrm{St}_L}.$$

If one group uses pipe radius and another uses diameter, the relation acquires the corresponding factor.

Illustrative Womersley numbers in canine vessels

The original tutorial used the following values for a canine cardiovascular system at a heart rate near $2\,\mathrm{Hz}$. Here the length scale in $\alpha = L\sqrt{\omega/\nu}$ is the vessel radius, $L = R$.

Vessel	α
Ascending aorta	13.2
Descending aorta	11.5
Abdominal aorta	8.0
Femoral artery	3.5
Carotid artery	4.4

Vessel	α
Arterioles	0.04
Capillaries	0.005
Venules	0.035
Inferior vena cava	8.8
Main pulmonary artery	15

These values illustrate the strong vessel-size dependence of α. They are not universal physiological constants: radius, frequency, and viscosity must accompany any reported value.

Values after W. W. Nichols and M. F. O'Rourke, *McDonald's Blood Flow in Arteries*, 5th ed. (Hodder Arnold, 2005).

Péclet number

For a scalar concentration c with molecular diffusivity D,

$$\frac{\partial c}{\partial t} + \boldsymbol{u} \cdot \nabla c = D \nabla^2 c.$$

With $c - c_0 = C c^*$ and the same L, U, and L/U scales,

$$\frac{\partial c^*}{\partial t^*} + \boldsymbol{u}^* \cdot \nabla^* c^* = \frac{1}{\mathrm{Pe}} \nabla^{*2} c^*, \qquad \boxed{\mathrm{Pe} = \frac{UL}{D}}.$$

$\mathrm{Pe} \ll 1$: diffusion dominates transport. $\mathrm{Pe} \gg 1$: advection dominates transport.

For mass transfer, $\mathrm{Pe} = \mathrm{Re}\,\mathrm{Sc}$; for heat transfer, $\mathrm{Pe} = \mathrm{Re}\,\mathrm{Pr}$, provided all groups use the same L and U.

Force scales for a 2-D model

Suppose $\mathcal{F}$ denotes a force resultant per unit out-of-plane depth, with units $\mathrm{N\,m^{-1}}$. For example, it may be the integral of boundary traction over a length of order L.

Normalization	Stress scale	Resultant scale
Inertial	ρU^2	$\mathcal{F}_I = \rho U^2 L$
Viscous	$\mu U/L$	$\mathcal{F}_V = \mu U$

$$\mathcal{F}_I^* = \frac{\mathcal{F}}{\rho U^2 L}, \qquad \mathcal{F}_V^* = \frac{\mathcal{F}}{\mu U}, \qquad \boxed{\mathcal{F}_V^* = \mathrm{Re}\,\mathcal{F}_I^*}.$$

A local force per unit boundary length and per unit depth has units $\mathrm{N\,m^{-2}}$ and scales with ρU^2 or $\mu U/L$. Begin with the force law and its measure.

Dimensionless bending coefficient in two dimensions

Under a two-dimensional per-unit-depth convention, let k_b have units

$$[k_b] = \mathrm{N\,m}.$$

The inertial and viscous normalizations are

$$\boxed{k_b^{(I)} = \frac{k_b}{\rho U^2 L^3}}, \qquad \boxed{k_b^{(V)} = \frac{k_b}{\mu U L^2}}.$$

They describe the same dimensional coefficient and satisfy

$$k_b^{(V)} = \mathrm{Re}\, k_b^{(I)}.$$

There is no mathematical switch at $\mathrm{Re} = 1$. Choose the normalization that exposes the balance of interest, then use it consistently.

Dimensionless stretching coefficient

Let k_s denote a force-like fiber coefficient per unit out-of-plane depth:

$$[k_s] = \mathrm{N\,m^{-1}}.$$

The corresponding nondimensional coefficients are

$$\boxed{k_s^{(I)} = \frac{k_s}{\rho U^2 L}}, \qquad \boxed{k_s^{(V)} = \frac{k_s}{\mu U}},$$

with

$$k_s^{(V)} = \mathrm{Re}\, k_s^{(I)}.$$

A point-spring stiffness and three-dimensional line, surface, or volume coefficients have different units. Begin with the exact force law and integration measure.

Viscosity and Reynolds-number matching

Because

$$\mathrm{Re} = \frac{\rho U L}{\mu} = \frac{U L}{\nu},$$

one can change Reynolds number by changing U, L, ρ, or μ.

- Varying μ is common in simulations and laboratory scale models.
- Changing viscosity may also change other material or transport parameters.
- Reporting only μ does not establish dynamic similarity; report Re and the chosen L and U.

When an oscillation or scalar field matters, matching Reynolds number alone is insufficient. Match the relevant Strouhal, Womersley, Péclet, and structural groups as well.

Flexural rigidity and effective stiffness

For a physical three-dimensional beam,

$$\boxed{B = EI},$$

where

- E is Young's modulus $[\mathrm{N\,m^{-2}}]$;
- I is a second moment of area $[\mathrm{m^4}]$;
- EI is flexural rigidity $[\mathrm{N\,m^2}]$.

Flexural rigidity is a material-and-section property. The force/displacement stiffness also depends on beam length, loading, and boundary conditions; commonly $K_{\mathrm{eff}} \sim EI/L_b^3$.

A 2-D model may use rigidity per unit out-of-plane depth, with units $\mathrm{N\,m}$. State that convention explicitly.

Approximate Young's modulus: compliant materials

Material	GPa	psi
Rubber, small strain	0.01–0.1	1,450–14,503
Low-density polyethylene	0.11–0.45	16,000–65,000
Diatom frustules, largely hydrated silica	0.35–2.77	50,000–400,000
PTFE	0.5	72,500
High-density polyethylene	0.8	116,000
Bacteriophage capsids	1–3	150,000–435,000
Polypropylene	1.5–2	218,000–290,000
Polyethylene terephthalate	2.2–2.7	320,000–390,000
Nylon	2–4	290,000–580,000
Polystyrene	3–3.5	440,000–510,000
Medium-density fiberboard	4	580,000
Oak, along the grain	11	1.60×10^6
Human cortical bone	14	2.03×10^6

These are order-of-magnitude reference values. Modulus depends on composition, orientation, hydration, strain, and test protocol.

Approximate Young's modulus: stiffer materials

Material	GPa	psi
Glass-reinforced polyester matrix	17.2	2.49×10^6
Aromatic peptide nanotubes	19–27	$2.76–3.92 \times 10^6$
High-strength concrete	30	4.35×10^6
Carbon-fiber-reinforced plastic, 50/50 biaxial	30–50	$4.35–7.25 \times 10^6$
Hemp fiber	35	5.08×10^6
Magnesium	45	6.53×10^6
Glass	50–90	$7.25–13.1 \times 10^6$
Flax fiber	58	8.41×10^6
Aluminum	69	10.0×10^6
Mother-of-pearl, largely calcium carbonate	70	10.2×10^6
Aramid	70.5–112.4	$10.2–16.3 \times 10^6$
Tooth enamel, largely calcium phosphate	83	12.0×10^6

Use a source appropriate to the actual material and loading direction before converting E into flexural rigidity EI.

Second moment of area

For a cross-sectional area A and an in-plane reference axis B,

$$\boxed{I_B = \iint_A r_\perp^2 \, dA},$$

where $r_\perp$ is the perpendicular distance from the axis to the area element dA.
For the Cartesian axes,

$$I_x = \iint_A y^2 \, dA, \qquad I_y = \iint_A x^2 \, dA.$$

The polar second moment about the perpendicular z axis is

$$J_z = \iint_A (x^2 + y^2) \, dA = I_x + I_y.$$

I_x and I_y control bending about the corresponding axes. J_z is a polar area moment and should not be confused with a mass moment of inertia.

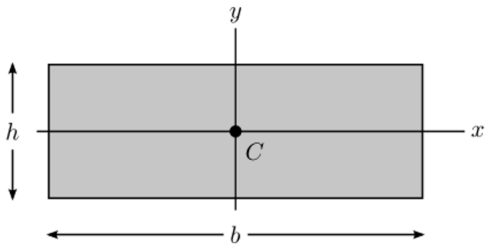
Rectangle with centroid at the origin

For width b in the x direction and height h in the y direction,

$$I_x = \int_{-b/2}^{b/2} \int_{-h/2}^{h/2} y^2 \, dy \, dx = \frac{bh^3}{12},$$

$$I_y = \int_{-h/2}^{h/2} \int_{-b/2}^{b/2} x^2 \, dx \, dy = \frac{b^3 h}{12},$$

$$J_z = I_x + I_y = \frac{bh}{12}(b^2 + h^2).$$



Rectangular cross-section of width b and height h with centroid C (schematic).

Converting structural parameters

1. Select and document one set of L and U.
2. Choose inertial or viscous normalization for the full coupled model.
3. Convert the desired dimensionless coefficient back to physical units.

For example, with inertial normalization,

$$k_{\text{spring}} = k^*_{\text{spring},I}\,\rho U^2 L, \qquad k_{\text{beam}} = k^*_{\text{beam},I}\,\rho U^2 L^3.$$

A coefficient does not determine deformation by itself. Geometry, loading, boundary conditions, and the constitutive law all matter, so report a sensitivity study rather than a universal "good" value.

Beam-stiffness starting estimate

With inertial normalization,

$$k_b = k_b^{(I)} \rho U^2 L^3.$$

The original course heuristic used $k_b^{(I)} \approx 10$ as a trial value that produced modest bending, roughly a few degrees, in a particular model.

- Use the same characteristic velocity and length as the fluid scaling; a peak velocity may be appropriate for a peak-load estimate. In practice: try $U_{\max}$ for the velocity scale and the boundary length for L; these are rough estimates and should be reported with the choice made.
- Compare the resulting k_b with a physical rigidity such as EI under the model's per-depth convention.
- Sweep $k_b^{(I)}$ and demonstrate convergence of the deformation or performance metric.

The numerical value 10 is a model-specific starting guess, not a universal relation between dimensionless stiffness and bending angle.

Physical load estimates

At moderate or high Reynolds number, estimate a total hydrodynamic load from a physical area:

$$F_{\mathrm{char}} \sim \frac{1}{2} C_D \rho U^2 A.$$

For a 2-D model per unit depth, use the corresponding projected length instead of area. At low Reynolds number, use a geometry-specific viscous drag law.

For a target displacement δ_{max},

$$K_{\mathrm{eff}} \sim \frac{F_{\mathrm{char}}}{\delta_{\mathsf{max}}}$$

provides an order-of-magnitude starting point.

Do not use the Eulerian grid spacing Δx as a physical area or mass unless you are explicitly constructing a discrete per-node quantity. Otherwise the physical model changes when the grid is refined.

Spring-stiffness starting estimate

If F_b is the characteristic load supported by a spring-like constraint and $\delta_{\max}$ is the largest acceptable displacement,

$$\boxed{K_{\text{eff}} \sim \frac{F_b}{\delta_{\max}}}.$$

Choose $\delta_{\max}$ from the physical tolerance, such as maximum displacement from a target point or resting length.

At moderate or high Reynolds number,

$$F_b \sim \tfrac{1}{2} C_D \rho U^2 A.$$

At low Reynolds number, use a geometry-specific viscous drag law rather than a universal grid-based estimate.

Convert the resulting stiffness using the units of the implemented force law. A node, line, surface, and volume spring require different coefficients.

Damping in a coupled fluid–structure system

For one linear degree of freedom,

$$m\ddot{x} + c\dot{x} + kx = 0, \qquad c_{\mathrm{crit}} = 2\sqrt{km}.$$

This formula is a useful analogy, but an immersed structure has distributed modes and frequency-dependent fluid loading.

- A massless boundary still experiences added mass and hydrodynamic damping.
- The effective mass is generally not $\rho(\Delta x)^d$.
- Select penalty damping with a modal estimate or solver-specific stability and convergence tests.

Report the damping law and demonstrate that the physical response and time-step restriction are adequately resolved.

Efficiency at steady swimming or flying

In a statistically steady or time-periodic state, mean kinetic energy does not grow, although it may oscillate instantaneously. Input power still balances drag and energizes the wake.

Constrained propulsor

A common definition is

$$\eta_{\mathrm{prop}} = \frac{TU}{P_{\mathrm{in}}},$$

where T is mean thrust.

Self-propelled body

Mean net force is zero. One alternative is

$$\eta_{\mathrm{QP}} = \frac{D_{\mathrm{tow}} U}{P_{\mathrm{in}}}.$$

Here D_{tow} is the drag from a matched towing test.

State the system boundary, useful-power definition, and treatment of negative actuator work; different efficiency measures answer different questions.

Cost of transport

If E_{in} is the input energy used to travel distance d,

$$C = \frac{E_{\mathrm{in}}}{d} = \frac{\overline{P}_{\mathrm{in}}}{\overline{U}} \quad [\mathrm{J\,m^{-1}} = \mathrm{N}].$$

Common normalizations are

$$C_m = \frac{E_{\mathrm{in}}}{md} = \frac{\overline{P}_{\mathrm{in}}}{m\overline{U}} \quad [\mathrm{J\,kg^{-1}\,m^{-1}}],$$

$$C^* = \frac{\overline{P}_{\mathrm{in}}}{mg\overline{U}} \quad \text{(dimensionless, weight-normalized)}.$$

For a distributed actuator, compute input power from the work rate using a stated sign convention, for example

$$P_{\mathrm{in}}(t) = \int_{\Gamma} \boldsymbol{f}_{\mathrm{act}} \cdot \boldsymbol{u}_b \, d\Gamma,$$

where $\boldsymbol{f}_{\mathrm{act}}$ is the actuator force density on the structure Γ and $\boldsymbol{u}_b$ is the boundary velocity. Average P_{in} over a whole number of cycles before forming C.

Nondimensionalization checklist

1. Define L, U, and every additional material or transport scale.
2. Substitute the scaled variables into the dimensional equations.
3. Choose one normalization that exposes the balance of interest.
4. Check every dimensionless coefficient by units and limiting cases.
5. Preserve the same definitions when comparing cases or redimensionalizing results.

Dynamic similarity requires matching every dimensionless group that materially affects the problem, not only Reynolds number.

References

- D. J. Acheson, *Elementary Fluid Dynamics*. Clarendon Press, 1990.
- S. Vogel, *Life in Moving Fluids*, 2nd ed. Princeton University Press, 1994.
- G. K. Taylor, R. L. Nudds, and A. L. R. Thomas, "Flying and swimming animals cruise at a Strouhal number tuned for high power efficiency," *Nature* 425 (2003), doi:10.1038/nature02000.
- J. R. Womersley, "Method for the calculation of velocity, rate of flow and viscous drag in arteries when the pressure gradient is known," *J. Physiol.* 127 (1955), doi:10.1113/jphysiol.1955.sp005276.
- A. J. Smits, "Undulatory and oscillatory swimming," *J. Fluid Mech.* 874 (2019), doi:10.1017/jfm.2019.284.
- C. S. Peskin, "The immersed boundary method," *Acta Numerica* 11 (2002), doi:10.1017/S0962492902000077.
- B. E. Griffith and X. Y. Luo, "Hybrid finite difference/finite element version of the immersed boundary method," *Int. J. Numer. Meth. Biomed. Engng.* 33 (2017), doi:10.1002/cnm.2888.